ALICE L WALTON SCHOOL OF MEDICINE

DNA Gyrase Inhibitors and Folic acid synthesis inhibitors–1 of 2: Sulfonamides

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Adapted from: Sulfonamides and Trimethoprim

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of DNA Gyrase Inhibitors and Folic acid synthesis inhibitors.
2. Compare and contrast the pharmacokinetics of DNA Gyrase Inhibitors ar
3. Identify the common drug interactions associated with DNA Gyrase and Folic acid synthesis inhibitors.
4. Recognize the common adverse reactions associated with DNA Gyrase and Folic acid synthesis inhibitors.

CASE CONNECTION

"It's another one of those urinary tract infections," reports CB, a 29-year-old female. "I had one last summer, and it was nasty. Now I've got the same burning and urgency." CB appears with normal vital signs and a normal physical, with no flank or abdominal tenderness. Urinalysis shows positive leukocyte esterase and WBCs. Urine culture was positive for *E. Coli* confirming the urinary tract infection (UTI). When CB was asked about her pregnancy status, she replied, "We are excited to welcome our first child, I still haven't completed my first trimester." She asked if she could go back on the medication that she had been given for her past UTIs. You look up her past medications and note that she was prescribed Bactrim(R) (sulfamethoxazole-Trimethoprim).

How will you explain your choice of antibiotics to CB? Consider your answer as you read, and we'll revisit CB at the end of the brick.

GO TO CONCLUSION ↓

What Are Sulfonamides and Trimethoprim?

What do eggs, grapef ir pungent, though very different, aromas come from sulfur compounds! Sulfur is essential for all life in many different processes, including the production and folding of proteins. The wide usage of sulfur compounds in nearly all lifeforms also means that sulfur-based compounds were likely applied as antimicrobials and have been used in one form or another to inhibit the growth of microbes for centuries. They are, in fact, the very first systemic antibiotics. Although penicillin usually gets the credit for being the first antibiotic, the sulfa drugs beat that "miracle drug" by more than a decade to launch the transformation of modern medicine.

These early systemic antibiotics, using the sulfonamide functional group (Figure 1), are still a major part of medical care today.

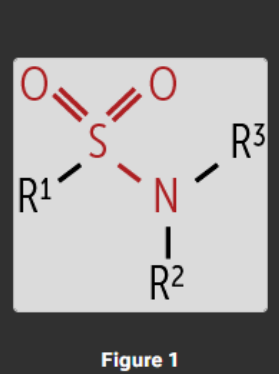


Figure 1

The sulfonamide functional group makes up a group of medications called sulfa drugs, which include many different classes of medications, such as diuretics, diabetes medications, and anti-inflammatory medications, among others. Here we will discuss their use as antimicrobials, specifically. They are often used with trimethoprim, a synergistic antibiotic acting on a similar system.

What is the core active structure of sulfonamide drugs?

How are Sulfonamide Classified? (LO#1)

Use of sulfonamide antibiotics, further simply called sulfonamides, has declined in recent years due to the development of resistance, although many others are still used in veterinary care. For human use, the primary sulfonamides are sulfadiazine and sulfamethoxazole (SMX). Sulfadiazine is classified as short-acting sulfonamides, while SMX is considered intermediate acting. All are available orally, often as combined medications. Sulfadiazine and SMX both have good systemic action, though SMX has better activity on gut pathogens, which we will discuss later. SMX is most often used in combination with trimethoprim (TMP), which acts on a different step in the same pathway to augment the activity of SMX. This general information is summarized in Table 1.

Table 1

Drug	Duration (Half-Life)	General Use	Combined With
Sulfadiazine	Short (10 h)	Systemic	Pyrimethamine
Sulfamethoxazole (SMX)	Intermediate (9-12 h)	Urinary tract infection, gut and skin infections, systemic	Trimethoprim (TMP)

CLINICAL CORRELATION

Long-acting and even ultra-long-acting sulfonamide antibiotics do exist, but these are rarely used in clinical practice due to limited efficacy data in humans, cost, side effects, or highly prevalent resistance. Some are still used in veterinary care, however.

How Do Sulfonamides Kill Bacteria?

Interestingly, sulfonamides do not actually kill bacteria—not on their own, at least. Sulfonamides are bacteriostatic. This means that they stop bacteria from growing and replicating, without actually actively killing them. In combination with other bacteriostatic medications, such as the combination of SMX with TMP, the combined bacteriostatic actions do become bactericidal, or lethal, to bacteria.

Sulfonamides work by acting as competitive inhibitors of the enzyme dihydropteroate synt ant job, creating a key precursor in the synthesis of tetrahydrofolic acid (THF), which is required to make DNA, RNA, and many proteins. While not truly folate inhibitors, sulfonamides are, effectively, inhibitors of nucleic acid synthesis (Figure 2).

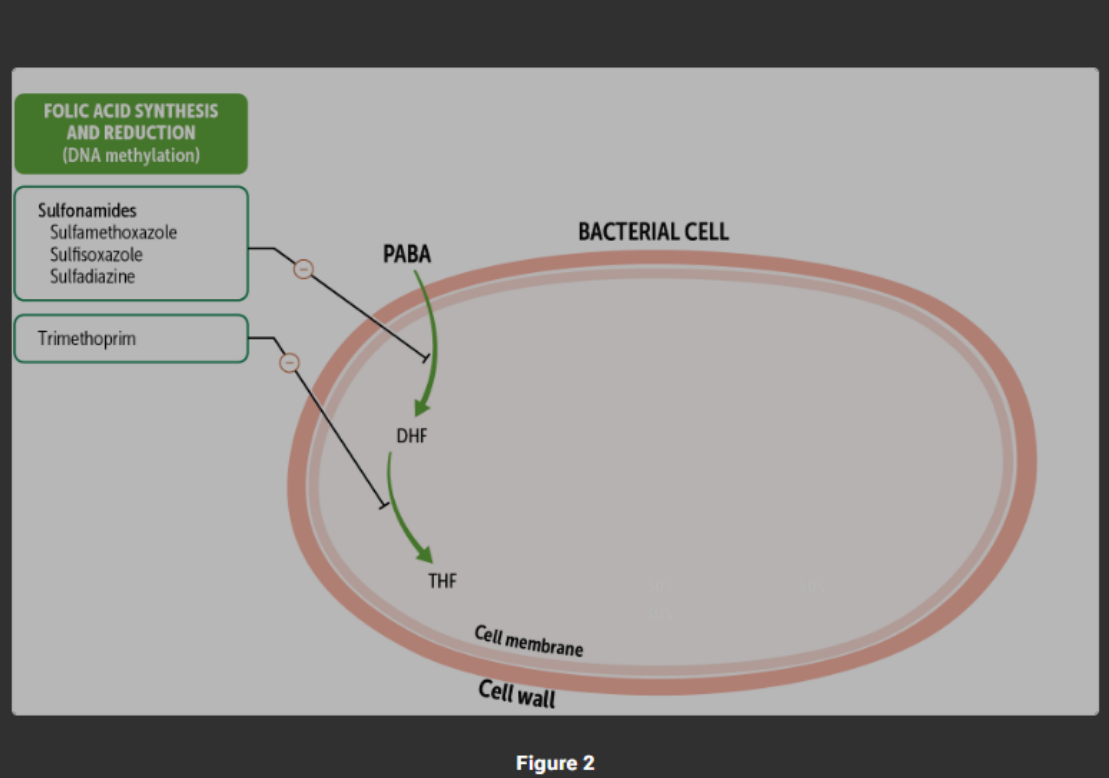


Figure 2

Credit: ©ScholarRx

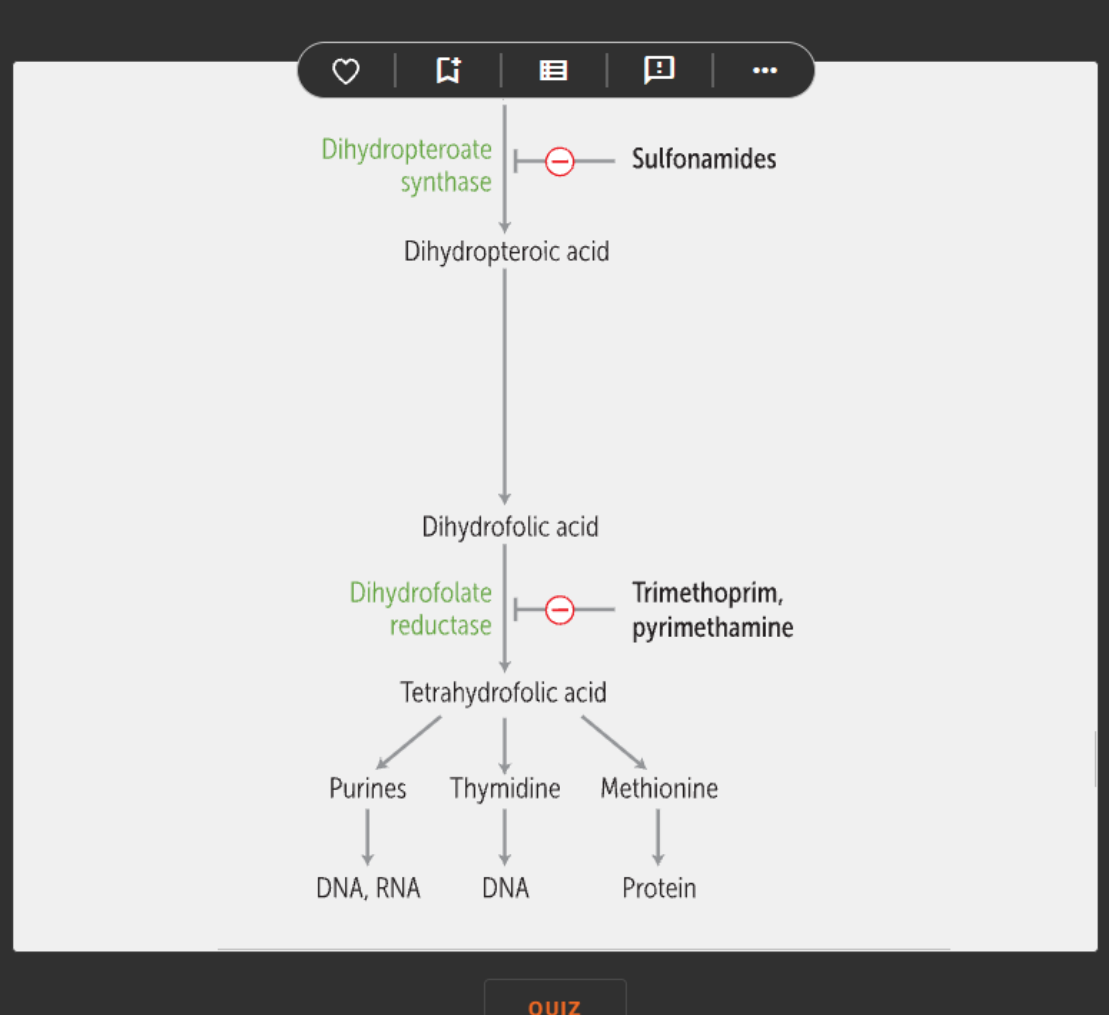
So, how is selective toxicity achieved by sulfonamides? We mammals cannot make folic acid; we must import it. We do not possess DHPS (see below under mechanism of action), the drug target. Instead, we have a folic acid transporter. Although some bacteria are auxotrophic for folic acid (meaning they do not produce it themselves and must ingest it), the majority must produce it themselves, leading t ainst most bacterial species.

Why is inhibition of dihydropteroate synthase an effective sulfonamide mechanism of action?

What is the mechanism of action of Sulfonamides?

So how does this really work? The process is shown in Figure 3, but we'll outline it step-by-step here:

1. Bacteria cells take in or produce para-aminobenzoic acid (PABA).
2. Bacterial DHPS uses PABA and pteridine to produce dihydropteroic acid.
3. Dihydropteroic acid is converted to dihydrofolic acid (DHF).
4. Bacterial dihydrofolate reductase converts DHF to tetrahydrofolic acid (THF).
5. THF is used to produce DNA, RNA, and proteins.



QUIZ

Figure 3

Credit: ©ScholarRx

Basically, sulfonamides are able to mimic PABA, inhibiting the action of DHPS in step 2. Without dihydropteroic acid, DHF cannot be produced to convert to THF, slowing the production of the DNA, RNA, and proteins that bacteria need to survive. This hints amides.

How do sulfonamides inhibit bacterial growth?

Synergy with Trimethoprim and Pyrimethamine

Figure 3 also helps explain the benefit of dual therapy of sulfonamides with TMP or pyrimethamine, two folate antagonists. Both of these medications inhibit the enzyme dihydrofolate reductase, which catalyzes the conversion of DHF to THF. TMP, in particular, has a stronger affinity for bacterial dihydrofolate reductase versus the human variant. This further reduces the overall amount of THF produced and, although TMP and pyrimethamine are bacteriostatic when used individually, the combination of these medications with their respective sulfonamide pairings produces synergistic bactericidal effects.

This combination helps overcome some of the resistance mechanisms we will discuss later, but to s ne-two punch that takes out a crucial system all at once. While one enzyme is relatively simple to develop a mutation in, mutating two crucial enzymes simultaneously is more unlikely, meaning the combination of a sulfonamide with a folate antagonist can help overcome resistance to one or the other medication.

Why is the combination of sulfonamides with TMP or pyrimethamine effective?

CLINICAL CORRELATION

It was previously thought that a combination of sulfonamides, called "triple sulfa preparations," would be more effective in treating certain diseases. Unfortunately, this theory did not pan out, with trials showing no good efficacy data. This outcome, however, makes sense given that all sulfonamides target the same enzyme.

What Defenses Do Microbes Have Against Sulfonamides?

So how does a bacterium defend against something attacking its nucleic acid and protein production? A few mechanisms have evolved, to be expected, given the near century that sulfonamides have been in use.

Sulfonamide resistance occurs through three primary mechanisms:

- Alteration of the bacterial DHPS enzyme
- Decreased uptake or increased efflux of sulfonamides
- Increased PABA synthesis

What are the key PK characteristics of Sulfonamides? (LO#2 and 3)

Most sulfonamides used today are taken orally, topically, and as parenteral formulations. All have very good oral absorption and good topical absorption. The two sulfonamides used commonly in the United States, sulfadiazine and SMX, are both metabolized in the liver, with SMX undergoing some metabolism by CYP2C9.

Drug interactions

- **Bleeding risk**: SMX/TMP is an inhibitor of CYP2C9 and can significantly increase INR, which can increase the potential for bleeding risk. Close monitoring of patients who are on these medications is advised.
- **Hyperkalemia**: sk of hyperkalemia with drugs such as

What enzyme does SMX/TMP inhibit, increasing INR and bleeding risk?

CLINICAL CORRELATION

As SMX undergoes partial metabolism by CYP2C9, it inhibits the metabolism of other drugs using this enzyme, most notably warfarin. This means that patients taking both warfarin and SMX will experience possibly increased anticoagulation effects from warfarin, thus increasing the risk of bleeding.

Distribution and Excretion

Following metabolism, active forms of these sulfonamides and their paired folate inhibitors are into most of the tissues and body fluids, including the urine, vaginal fluid, and middle ear fluids. Both sulfadiazine and SMX have good cerebrospinal fluid penetration.

Both SMX and sulfadiazine are excreted primarily through the urine, with a significant amount of unmetabolized sulfonamide being found in the urine. This allows sulfonamides to have good efficacy for the treatment of UTIs.

How are the sulfonamide drugs primarily excreted out of the body?

What Adverse Drug Reactions can Sulfonamides Cause? (LO#4)

- The common adverse reactions are **GI upset** (nausea/vomiting) and **skin** (rash and pruritus).
- **Hyperkalemia** can be a potentially life-threatening condition in the elderly and in pat t can increase the hyperkalemia risk
- Mammals, including humans, take in much of their folate through their diet and do not express the DHPS enzyme. However, TMP inhibits the human dihydrofolate reductase enzyme. In patients with **folate deficiency**, supplementation with folic acid is recommended.
- The use of SMX/TMP can result in **crystalluria**; therefore, patients should be recommended to drink at least 8 oz when they take the tablet. This helps to solubilize the drug.
- More rare reactions occur in patients with **glucose-6-phosphate dehydrogenase (G6PD) deficiency** (identified using a positive Coombs test). In these patients, exposure to sulfonamides can cause a rapid acute hemolytic anemia.
- Lastly, and while very rare, sulfonamides may induce **Stevens-Johnson syndrome (SJS)** and its deadlier sibling **toxic epidermal necrolysis (TEN).** These severe autoimmune skin rashes (Figure 5a and Figure 5b) exist on a spectrum based on how much of the body is affected but should any patient taking sulfonamides begin to experience a rash, the offending medication should be discontinued, as SJS and TEN can be fatal.

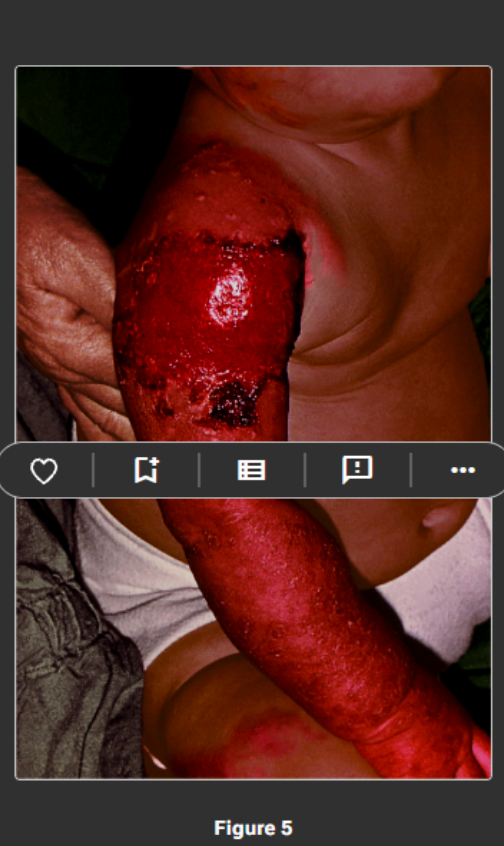
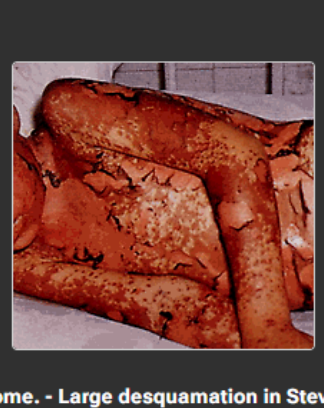


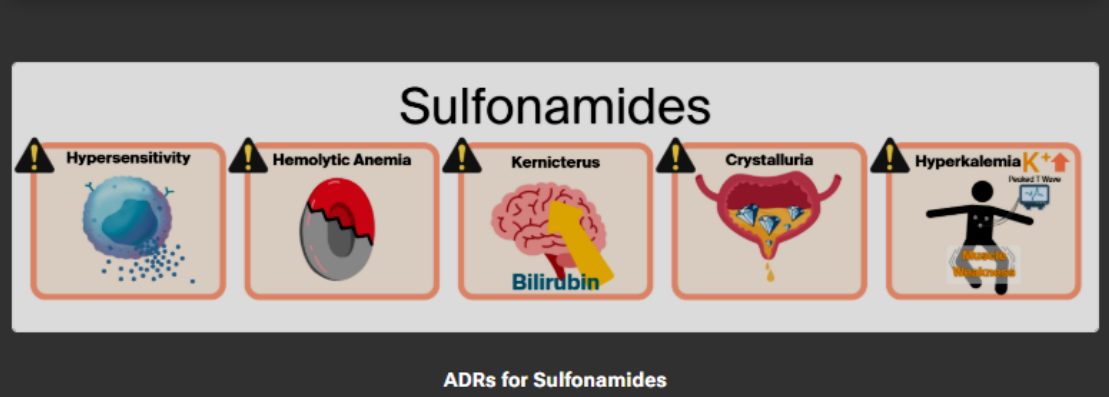
Figure 5



Stevens-Johnson syndrome. - Large desquamation in Stevens-Johnson syndrome.

Pregnancy – During certain stages of pregnancy,TMP-SMX should be avoided, if possible, as it may be associated with an increased risk of adverse events. In general, TMP-SMX should be avoided in the first trimester of pregnancy since trimethoprim may interfere with folate metabolism and increase the risk of neural tube defects. TMP-SMX should also be avoided, if possible, in the last month of pregnancy, due to the ability of sulfonamides to displace bilirubin bound to serum albumin, thereby increasing free unconjugated bilirubin levels that could increase tl

Why should TMP-SMX be avoided in the first trimester of pregnancy?



ADRs for Sulfonamides

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to CB, how do you explain your choice of antibiotics?

You explain to CB, " ". You explain to CB that SMX-TMP is contraindicated in the first trimester of pregnancy. The use of SMX-TMP can cause fetal damage, and hence, other safer antibiotics should be considered. You also encourage her to read about UTIs and her medications but not to worry too much about the more severe adverse effects because they truly are not so common.

Summary

- Sulfonamides are the first systemic antibiotics and include many different classes of medications, such as diuretics, diabetes medications, and anti-inflammatory medications.
- The core active structure of sulfonamide drugs is the sulfonamide functional group.
- Sulfonamides work by acting as competitive inhibitors of the enzyme dihydropteroate synthase (DHPS), preventing nucleic acid synthesis.
- Sulfonamides are often used in combination with trimethoprim (TMP), a synergistic antibiotic acting on a similar system.
- Resistance to sulfonamides can occur through alteration of the bacterial DHPS enzyme, decreased uptake or increased efflux of sulfonamides, or increased PABA synthesis.
- Adverse reactions to sulfonamides can include gastrointestinal upset, skin reactions, hyperkalemia, and in rare cases, Stevens-Johnson syndrome or toxic epidermal necrolysis.

Review Questions

1. How do sulfonamides inhibit bacterial growth?

A. ○ Competetive inhibition DNA polymerase
B. ● Competitive inhibition of DHPS
C. ○ Inhibit RNA polymerase
D. ○ Inhibition of DNA polymerase

Hide Explanation

The correct answer is competitive inhibition of DHPS (B). Sulfonamides act as competitive inhibitors of the enzyme dihydropteroate synthase (DHPS), preventing the synthesis of tetrahydrofolic acid (THF). Inhibition of DNA polymerase (A) and RNA polymerase (C) are mechanisms of other antibiotics. Sulfonamides do not inhibit DNA polymerase directly (D).

2. Which condition can SMX/TMP use increase the risk of?

A. ○ Hypokalemia
B. ● Hyperkalemia
C. ○ Hypoalbuminemia
D. ○ Hypocalc

Hide Explanation

The correct answer is hyperkalemia (B). SMX/TMP can increase the risk of hyperkalemia, especially when used with ACEi/ARBs and NSAIDs. Hypokalemia (A) is incorrect as SMX/TMP does not typically cause low potassium levels. Hypoalbuminemia (C) is incorrect. SMX/TMP can increase albumin levels. Hypercalcemia (D) is also incorrect, as SMX/TMP does not affect calcium levels.

3. Which CYP interactions increase INR (bleeding risk) with the concurrent use of SMX/TMP with warfarin?

A. ● CYP2C9 inhibition
B. ○ CYP2C19 induction
C. ○ CYP3A4 inhibition
D. ○ CYP 2C9 induction

Hide Explanation

The correct answer is CYP2C9 inhibition (A). SMX/TMP inhibits CYP2C9, leading to increased INR and bleeding risk. Warfarin is metabolized by 2C9; hence, inhibition of the enzyme can increase bleeding risk. CYP2C19 induction will decrease INR. Warfarin is not a maj nce (C) is incorrect. Choice (D) is incorrect since warfarin is not a CYP 2C9 inducer.

4. Drinking 'plenty of water' (at least 8oz) is recommended when taking SMX/TMP to avoid

A. ○ Anuria
B. ● Crystalluria
C. ○ Dysuria
D. ○ Polyuria

Hide Explanation

The correct answer is crystalluria (B). SMX/TMP has been shown to cause crystalluria. Drinking plenty of water with the drug helps in dissolving the salt and thus helps reduce the incidence. Choice (A), anuria does not develop with the use of SMX/TMP. Choice's dysuria (C), and polyuria (D) are not associated with the use of SMX/TMP.

5. SMX/TMP would cover infections from which of these organisms?

A. ○ *B. fragillis*
B. ○ *C. botulin*
C. ○ *C. dificile*
D. ● *K. pneumoniae*

Hide Explanation

The correct answer is *K. pneumoniae (D)*. SMX/TMP has broad coverage against many aerobic gram-positive and -negative species like *K. pneumoniae*. *B. fragillis* (A) is incorrect because it is an anerobic organism. *C. botulinum* (B) and *C. dificile* (C) are both incorrect because *Clostridium* is an anaerobic organism as well, which is not covered by SMX/TMP.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action and mechanism of resistance of DNA Gyrase Inhibitors and Folic acid synthesis inhibitors.~~

02 ☑ ~~Compa~~ ~~Inhibito~~ ~~NA Gyrase~~

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Objective Confidence: 4/4